

Kaden Seto

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EDUCATION

University of Toronto

Bachelor of Applied Science & Engineering (BASc) in Engineering Science + PEY Co-op

Toronto, ON

Sep 2024 - 2029

TECHNICAL SKILLS

Languages: Python, C#, C/C++, JavaScript/TypeScript, Java, HTML/CSS, MATLAB

Developer Tools: Git, Visual Studio Code, Unity, Node.js, Android Studio

Libraries: PyTorch, Gymnasium & SB3, Unity ML-Agents, Keras, NumPy, pandas, OpenCV, scikit-learn

ML: RL (Deep RL, MARL), RNN/LSTM, NODEs, Computer Vision, Recommender Systems, Transformers, GANs

EXPERIENCE

University of Toronto Machine Intelligence Student Team | [Website](#)

Toronto, ON

Vice President, Academics (May 2025 - present)

Nov 2024 - present

- Managing the Academics department and initiatives, encompassing **12 directors & leads** across 5 Academic workshops, events, and programs (in collaboration with industry partners including **IBM, AMD Schola, Riot Games**) supporting **3000+** students (**North America's largest undergraduate ML/AI team**)

Reinforcement Learning Team Lead (Nov 2024 - Mar 2025) | [Github](#)

- Directed a team of **6 developers** to build **AI² (Canada's largest RL tournament)**, for **400+** participants
- Developed a custom **multi-agent RL self-play training process** supporting custom **PyTorch + SB3** agent implementations and a custom platform-fighting game environment using **Gym, Pygame & Pymunk**
- First-authored** a paper at **CUCAI (Canada's largest undergrad AI conference)**

BMO Financial Group

Toronto, ON

Software Developer Intern

May 2025 - Aug 2025

- Developed Python scripts for automating environment syncing and AWS CDK Stack deployment, **reducing process time by 2+ hours**

PROJECTS

Continuum | [Github](#)

Oct 2025 - present

- Researching & developing a dimension-agnostic deep RL framework in **PyTorch + SB3** for **physics-informed RL**, combining model-free **RL algorithms** (e.g. PPO, SAC), **Neural ODEs**, and **Autoencoders (AEs)** to introduce a custom neural network architecture for solving **Gym/MuJoCo** environments.
- Achieving **200-300% improvement** in reward performance compared to regular RL algorithms.

NeuroStrike | [Github](#)

May 2025 - Aug 2025

- Engineered a multi-agent RL framework for **training martial arts AI agents** to spar in a **custom-made Unity3D environment** simulation using **self-play recurrent PPO (LSTM + PPO)**.
- Engineered hyperparameters, architectures, reward functions, and observation spaces to optimize policy behaviour, achieving **~72% improvement** in strategic behaviours and **~94% increase** in cumulative reward.

Aegis | [Github](#)

Jan 2025 - Apr 2025

- Collaborated with clients (Advocacy for Respect for Cyclists group) to engineer a **Cyclist Detection and Arduino flashing LED** system aimed to prevent right-hooking cyclist incidents in Toronto
- Trained a custom object detection model using **YOLOv11** architecture, achieving a **mAP50-95 of 84.6%**, average **50ms inference time**, and detections up to **3.2m** on a phone camera
- Implemented **Swift-SRGAN** in **PyTorch** for real-time super-resolution to enhance detection.

QuantNet | [Github](#)

Oct 2024 - Feb 2025

- Implemented a hybrid **LSTM & MLP neural network framework from scratch** in **C++** with **0 external dependencies** (i.e. methods such as back-propagation and Adam made from scratch) for investment analysis
- Implemented a **feature engineering framework from scratch** by researching key **investment analysis indicators** to enhance model quality and performance (e.g. features such as RSI, ADX)